1. (b)

$$y - x\frac{dy}{dx} = a\left(y^2 + \frac{dy}{dx}\right)$$

$$\Rightarrow y - ay^2 = a\frac{dy}{dx} + x\frac{dy}{dx}$$

$$\Rightarrow y(1 - ay) = (a + x)\frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{(a+x)} = \frac{dy}{y(1-ay)}$$

On integrating both sides, we get

$$\int\frac{dx}{(a+x)} = \int\frac{dy}{y(1-ay)}$$

$$\Rightarrow \log(a + x) = \int\left[\frac{1}{y} + \frac{a}{(1-ay)}\right]dx$$

$$\Rightarrow \log(a + x) = \log y + \frac{a\log(1-ay)}{-a} + \log c$$

$$\Rightarrow \log(a + x) = \log y - \log(1 - ay) + \log c$$

$$\Rightarrow \log(x + a)(1 - ay) = \log cy$$

$$\Rightarrow (x + a)(1 - ay) = cy$$

2. (a)

Given equation may be re-written as

$$\frac{y}{x}\cdot\frac{dy}{dx} = \left(\frac{y}{x}\right)^2 + \frac{\phi((y/x)^2)}{\phi'((y/x)^2)} \quad .....(i)$$

Let $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$ and $\frac{y}{x} = v$

$\therefore$ From (i), $v\left(v + x\frac{dv}{dx}\right) = v^2 + \frac{\phi(v^2)}{\phi'(v^2)} \Rightarrow vx\frac{dv}{dx} = \frac{\phi(v^2)}{\phi'(v^2)}$

$$\Rightarrow \frac{\phi'(v^2)(2v\,dv)}{\phi(v^2)} = 2\frac{dx}{x}$$

Integrating, $\ln(\phi(v^2)) = 2\ln x + \ln c \Rightarrow \phi(v^2) = cx^2$

$\therefore \phi(y^2/x^2) = cx^2$

3. (a)

Re-writing the given equation,

$$2xy^2dx + ye^xdx = e^xdy \Rightarrow 2x\,dx + \frac{ye^xdx - e^xdy}{y^2} = 0$$

$$\Rightarrow d(x^2) + d\left(\frac{e^x}{y}\right) = 0$$

Integrating, $x^2 + \frac{e^x}{y} = c$

$\therefore yx^2 + e^x = cy$

4. (d)

For $\phi(x) = y$, $y' = 1 + y^2 \Rightarrow \frac{dy}{dx} = 1 + y^2 \Rightarrow \int\frac{dy}{1+y^2} = \int dx$

$\Rightarrow \tan^{-1}y = x + c$

$\therefore$ $y = \tan(x + c)$

i.e., $\phi(x) = \tan(x + c)$

As $y(0) = 0$, $0 = \tan c$ and $y(\pi) = 0 \Rightarrow 0 = \tan(\pi + c) = \tan c$

$\therefore c = 0$

$\therefore$ $\phi(x) = y = \tan x$.

But $\tan x$ is not continuous in $(0, \pi)$

Since $\tan\frac{\pi}{2}$ is not defined.

Hence there exists not a function satisfying the given condition.

5. **(a)**

The equation of family is $y = Ax^2 + Bx + C$, where A, B, C are arbitrary constants. Hence, $\frac{d^3y}{dx^3} = 0$

6. (d)

We have, $y = (C_1 + C_2)\sin(x + C_3) - C_4e^{x+C_5}$

$\Rightarrow$ $y = C_6\sin(x + C_3) - C_4e^{C_5}.e^x$, where $C_6 = C_1 + C_2$

$\Rightarrow$ $y = C_6\sin(x + C_3) - C_7e^x$, where $C_4e^{C_5} = C_7$ Clearly, the above relation contains three arbitrary constants. So, the order of the differential equation satisfying it is 3.

7. **(d)**

Solving the homogeneous equation, by using

$y = vx$, we find the solution $x + y = c(x^2 + y^2)$ $y(-1) = 1 \Rightarrow x + y = 0$ which is a straight line.

8. **(b)**

[K is constant of proportionality] Let the curve be $y = f(x)$. The equation of tangent at any point (x, y) is given by $Y - y = f'(x)(X - x)$. So the portion of the axis of x which is cut off between the origin and the tangent at any point is obtained by putting $Y = 0$. Therefore,

$x - \frac{y}{f'(x)} = Ky \Rightarrow x - y\frac{dx}{dy} = Ky \Rightarrow \frac{dx}{dy} - \frac{x}{y} = -K$

which is a linear equation in x, so its integrating factor is $e^{-\int(1/y)dy} = y^{-1}$. Therefore, multiplying by y^{-1}, we have

$\frac{d}{dy}(xy^{-1}) = -Ky^{-1} \Rightarrow xy^{-1} = -K\log y + C$

$\Rightarrow x = y(C - K\log y)$

where C is arbitrary constant

9. **(c)**

We can write $y = A\cos(x + B) - Ce^x$

where $A = c_1 + c_2$, $B = c_3$ and $C = c_4e^{c_5}$

$\frac{dy}{dx} = -A\sin(x + B) - Ce^x$

$\Rightarrow \frac{d^2y}{dx^2} = -A\cos(x + B) - Ce^x$

$\Rightarrow \frac{d^2y}{dx^2} + y = -2Ce^x$

$\Rightarrow \frac{d^3y}{dx^3} + \frac{dy}{dx} = -2Ce^x = \frac{d^2y}{dx^2} + y$

$\Rightarrow \frac{d^3y}{dx^3} - \frac{d^2y}{dx^2} + \frac{dy}{dx} - y = 0$

Which is a differential equation of degree 1.

Hence (C) is the correct answer

10. (d)

Since the equation is not a polynomial in all the differential coefficients so the degree is not defined.

11. (b)

The slope of the ray = $\frac{y}{x}$

Slope of the normal at P (x, y) = - $\frac{dx}{dy}$

$\Rightarrow \frac{\frac{y}{x}+\frac{dx}{dy}}{1-\frac{y}{x}\frac{dx}{dy}}$

$\Rightarrow \frac{y}{x}\frac{dx}{dy} + 1 = - 1 + \frac{y}{x}\frac{dx}{dy} \Rightarrow y\left(\frac{dy}{dx}\right)^2 + 2x\frac{dy}{dx} = y$ is differential equation of the curve which is satisfied by $y^2 = 2x + 1$.

Hence (B) is the correct answer

12. (a)

The given equation is

$t = 1 + (ty)\left(\frac{dy}{dt}\right) + \frac{(ty)^2}{2!}\left(\frac{dy}{dt}\right)^2 + \ldots \infty \Rightarrow t = e^{ty\left(\frac{dy}{dt}\right)} \Rightarrow \log t = ty\frac{dy}{dt}$

$\Rightarrow y\,dy = \frac{\log t}{t}\,dt$

On integration

$\frac{y^2}{2} = \frac{(\log t)^2}{2} + k$

$\Rightarrow y = \pm\sqrt{(\log t)^2 + 2k}$

$\Rightarrow y = \pm\sqrt{(\log t)^2 + c}$

13. (b)

$x = r\cos\theta,\ y = \sin\theta,\ \frac{dr}{|-r^2|} = d\theta \to \sin^{-1}r + \theta + \alpha$

$\Rightarrow r = \sin(\theta + \phi)$

$r^2 = y\cos\alpha + x\sin\alpha$

$x^2 + y^2 - x\sin\alpha - y\cos\alpha = 0$

$\therefore$ radius = ½

14. (c)

$\frac{dy}{dx} + \frac{x}{1-x^2}y = \frac{5x}{1-x^2}$ whose solution is

$y.e^{\int\frac{x^2}{1-x^2}dx} = \int\left(\frac{5x}{1-x^2}\right)$ I.F. dx + c

15. (b)

$\frac{x^2(x\,dx + y\,dy)}{\sqrt{x^2+y^2}} = ydx - x\,dy$

Or $\int\frac{1}{2}\frac{d(x^2+y^2)}{\sqrt{x^2+y^2}} = \int\frac{ydx - xdy}{x^2}$

$\Rightarrow \frac{1}{2}\, 2.\, \sqrt{x^2+y^2} = -\frac{y}{x} + C$

16. (d)

We have, $(x + 2y^3)\frac{dy}{dx} = y \Rightarrow y\frac{dx}{dy} = x + 2y^3$

$\Rightarrow \frac{dx}{dy} - \frac{1}{y}x = 2y^2,$

Which is a linear equation, if we take x as the dependent variable.

I.F. $= e^{\int p dy} = e^{-\int \frac{1}{y}dy} = e^{-\log y} = e^{\log\left(\frac{1}{y}\right)} = \frac{1}{y}.$

$\therefore$ The solution is $x.\frac{1}{y} = \int 2y^2 . \frac{1}{y}\, dy + c$

$\Rightarrow x . \frac{1}{y} = y^2 + c \Rightarrow x = y\,(c + y^2).$

Hence (D) is the correct answer.

17. (a)

We have, $(xy - x^2)\frac{dy}{dx} = y^2 \Rightarrow y^2\frac{dx}{dy} = xy - x^2$

$\Rightarrow \frac{1}{x^2}\frac{dx}{dy} - \frac{1}{x} . \frac{1}{y} = \frac{-1}{y^2}$

Let $\frac{1}{x} = V \Rightarrow -\frac{1}{x^2}\frac{dx}{dy} = \frac{dV}{dy}$, we obtain

$\frac{dV}{dy} + \frac{V}{y} = \frac{1}{y^2}$, which is linear.

I. F. $= e^{\int \frac{1}{y}dy} = e^{\log y} = y.$

$\therefore$ The solution is

$Vy = \int \frac{1}{y^2} . y\, dy + c$

$\Rightarrow \frac{y}{x} = \log y + c$

$\Rightarrow y = x\,(\log y + c).$

This passes through the point (-1, 1).

$\therefore 1 = -1\,(\log 1 + c) \Rightarrow c = -1.$

Hence (A) is the correct answer

18. (d)

$\frac{d^3y}{dx^3}\left(\frac{d^2y}{dx^2}\right)^5 + 4\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{d^3y}{dx^3}\right)^2$

$= (x^2-1)\frac{d^3y}{dx^3}$

order = 3 = m

Degree = 2 = n

19. (c)

$\left(\frac{d^3y}{dx^3}\right)\left(\frac{d^2y}{dx^2}\right)^5 + 4\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{d^3y}{dx^3}\right)^2$

$= (x^2 - 1)\left(\frac{d^3y}{dx^3}\right)$

order = 3, degree = 2

20. (d)

only one parameter c, eliminate it.